Naturalistic Induction Heads: How Data Distribution Shapes Pattern Completion in Transformers

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Abstract

Induction heads—attention heads that complete patterns of the form $[A][B]\ldots[A] \rightarrow B$—are considered fundamental to in-context learning in transformers. Yet they have been primarily identified and studied using random repeated token sequences, an artificial setup far from the natural language these models were trained on. We ask: do the same heads that perform induction on random data also do so on naturalistic text, or are there "hidden" induction mechanisms that only activate on training-distribution-like inputs? We measure per-head induction scores on both random repeated sequences and OPENWEBTEXT passages in GPT-2-SMALL (124M parameters), complemented by output-value (OV) copying analysis and causal ablation experiments. We find that the rank ordering of heads is highly conserved across conditions (Spearman $\rho = 0.84$, top-5 overlap = 80%), ruling out the existence of naturalistic-only induction heads in the standard sense. However, induction scores are 10–40$\times$ lower on natural text, and late-layer heads (layers 9–11) show dramatically *stronger* copying behavior on natural text than on random data—up to 29$\times$ higher OV copying scores. Ablating the 21 random-detected induction heads increases natural-text loss by 76.6%. Our results show that standard random-sequence detection identifies the right heads but underestimates their natural-language capabilities: the same induction heads implement a richer, distribution-sensitive form of pattern completion on naturalistic data.

1 Introduction

Induction heads are among the most well-characterized circuits in transformer language models. First identified by Elhage et al. [2021] and extensively studied by Olsson et al. [2022], these attention heads implement a simple but powerful pattern: given a sequence $[A][B]\ldots[A]$, they predict B by attending to the token following the previous occurrence of A. This mechanism is believed to underlie much of in-context learning (ICL), the ability of language models to adapt their predictions based on context without weight updates.

But how do we know which heads are induction heads? The standard detection protocol Olsson et al. [2022] feeds the model random repeated token sequences—e.g., [`cat`][`dog`]...[`cat`]—and measures whether each head attends to the position after the first occurrence of the repeated token. This artificial setup provides a clean, controlled signal, but it is far from the natural language on which these models were trained. A head that scores highly on random repeated sequences may behave quite differently when processing real text, where attention must be distributed across many semantically and syntactically relevant positions rather than concentrated on a single induction target.

We hypothesize that this gap between detection conditions and deployment conditions may cause the standard protocol to miss important aspects of how induction mechanisms operate on naturalistic data. Specifically, there may exist heads that perform pattern completion on natural text but remain

silent on random sequences, or conversely, heads identified as strong induction heads on random data may contribute to natural language processing through mechanisms that the standard induction score fails to capture.

To test this, we conduct a systematic comparison of induction head behavior on random versus naturalistic data in GPT-2-SMALL (124M parameters). We apply the standard induction score Olsson et al. [2022] identically to both random repeated sequences and OPENWEBTEXT passages, then complement this attention-pattern analysis with output-value (OV) copying scores that measure what each head *does* with the information it attends to. We further validate our findings through ablation experiments that measure the causal contribution of identified heads to natural language processing.

Our main findings are:

- We show that induction heads identified on random data are the *same* heads that show the strongest induction behavior on natural text (Spearman $\rho = 0.84$, top-5 overlap = 80%), ruling out the existence of hidden naturalistic-only induction heads.
- We demonstrate that induction scores are 10–40$\times$ lower on natural text despite similar token repeat rates (41% vs. 50%), revealing that the standard metric dramatically overstates induction pattern strength under naturalistic conditions.
- We discover that late-layer heads (layers 9–11) exhibit up to 29$\times$ stronger OV copying on natural text than on random data, indicating distribution-sensitive output circuits that the attention-pattern-based detection protocol underestimates.
- We confirm through ablation that the 21 random-detected induction heads are causally important for natural text processing, increasing loss by 76.6% when removed.

The rest of this paper is organized as follows. Section 2 reviews prior work on induction heads and in-context learning mechanisms. Section 3 describes our experimental methodology. Section 4 presents our results, and section 5 discusses their implications and limitations. We conclude in section 6.

2 Related Work

Induction heads and transformer circuits. Elhage et al. [2021] introduced a mathematical framework for understanding transformer circuits, identifying induction heads as a two-head composition: a "previous token" head in an early layer composes with an induction head in a later layer to implement the $[A][B] \ldots [A] \rightarrow B$ pattern. Olsson et al. [2022] scaled this analysis across 34 models (up to 13B parameters), demonstrating that induction heads form during a sharp phase transition early in training that coincides with a dramatic improvement in in-context learning ability. Their detection methodology—measuring attention patterns on random repeated sequences—has become the standard approach in the field. Singh et al. [2024] decomposed induction head formation into three subcircuits (previous token attention, QK matching, and copying) and showed that their multiplicative interaction produces the phase change. Unlike these works, which primarily characterize induction heads using random sequences, we directly compare head behavior across random and naturalistic conditions.

Data distribution and mechanism formation. Several recent works have examined how training data properties affect the mechanisms transformers learn. Bietti et al. [2023] studied induction head emergence through an associative memory lens using quasi-naturalistic bigram statistics, finding that data distribution properties significantly affect formation speed. Kawata et al. [2025] proved that data diversity determines whether transformers learn induction heads or positional shortcuts: sufficient diversity yields generalizable induction heads, while low diversity yields brittle positional heuristics. Edelman et al. [2024] showed that induction heads on Markov chain data can compute calibrated conditional probabilities rather than performing deterministic copying, suggesting that the underlying mechanism is richer than the standard detection metric implies. Our work complements these theoretical and synthetic-data studies by empirically measuring how a trained model's induction behavior differs between random and naturalistic inputs.

Semantic induction and beyond-copying mechanisms. Most relevant to our work, Ren et al. [2024] discovered "semantic induction heads" in InternLM2-1.8B that encode syntactic dependencies and knowledge graph relations rather than performing token-level copying. They found that these heads correlate with the emergence of pattern discovery abilities during training, and that some

pure copying heads' scores actually *decrease* as semantic capabilities emerge. Our work differs in two key respects: (1) we systematically compare the *same* heads across random and naturalistic conditions rather than identifying a new class of heads, and (2) we decompose the discrepancy into attention-pattern (QK) versus output-circuit (OV) contributions, revealing that late-layer heads have distribution-sensitive copying circuits.

3 Methodology

We study induction head behavior in GPT-2-SMALL (124M parameters, 12 layers, 12 heads per layer) using TRANSFORMERLENS Nanda and Bloom [2022]. Our experimental protocol consists of four complementary analyses: standard induction score comparison, fuzzy (embedding-similarity) induction, OV copying analysis, and causal ablation.

3.1 Data Construction

Random repeated sequences. Following Olsson et al. [2022], we construct 50 sequences of 257 tokens each: a BOS token followed by 128 random tokens sampled uniformly from the vocabulary, then the same 128 tokens repeated. By construction, 50.0% of tokens in the second half are exact repeats of tokens in the first half.

Naturalistic sequences. We sample 50 passages from OPENWEBTEXT (articles with $\geq$500 characters), tokenize with GPT-2's BPE tokenizer, prepend BOS, and truncate to 257 tokens. Natural text has a token repeat rate of 41.1% $\pm$ 5.4%, surprisingly close to the 50% rate in random sequences. This means that differences in induction scores between conditions are not primarily driven by fewer opportunities for induction on natural text.

3.2 Induction Score

For each head h at layer l, we compute the **induction score** on a sequence $\boldsymbol{x} = (x_1, \ldots, x_T)$ as follows. Let $\alpha_{i,j}^{(l,h)}$ denote the attention weight from position i to position j for head h in layer l. For each position i where $x_i = x_k$ for some $k < i$ (a repeated token), the induction score contribution is $\alpha_{i,k+1}^{(l,h)}$—the attention paid to the position *after* the previous occurrence. We average this quantity over all repeated-token positions and over all sequences:

$$\mathrm{IS}(l,h) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{1}{|R_s|} \sum_{i \in R_s} \alpha_{i,\mathrm{prev}(i)+1}^{(l,h)} \tag{1}$$

where $\mathcal{S}$ is the set of sequences, R_s is the set of positions with a repeated token in sequence s, and prev(i) returns the most recent prior position where the same token appeared. This score is computed identically on both random and naturalistic data.

3.3 Fuzzy Induction Score

To test whether naturalistic induction operates over semantically similar (rather than identical) tokens, we relax the exact-match requirement. We compute cosine similarity between token embeddings and consider position k a "match" for position i if $\cos(\boldsymbol{e}_{x_i}, \boldsymbol{e}_{x_k}) > 0.8$, where $\boldsymbol{e}_x$ denotes the embedding of token x. The fuzzy induction score is then computed as in equation 1 but over these similarity-based matches.

3.4 OV Copying Score

The induction score measures *where* a head attends (the QK circuit). To measure *what* it does with that information (the OV circuit), we compute a copying score. For each head on each sequence, we: (1) identify the most-attended position j^* (excluding BOS and self-attention), (2) compute the head's output $\boldsymbol{z}^{(l,h)}\boldsymbol{W}_O^{(l,h)}$, (3) project through the unembedding matrix $\boldsymbol{W}_U$ to obtain logits, and (4) measure whether the logit of the attended token x_{j^*} exceeds the mean logit:

$$\mathrm{CS}(l,h) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left((\boldsymbol{z}^{(l,h)}\boldsymbol{W}_O^{(l,h)}\boldsymbol{W}_U)_{x_{j^*}} - \overline{\mathrm{logits}} \right) \tag{2}$$

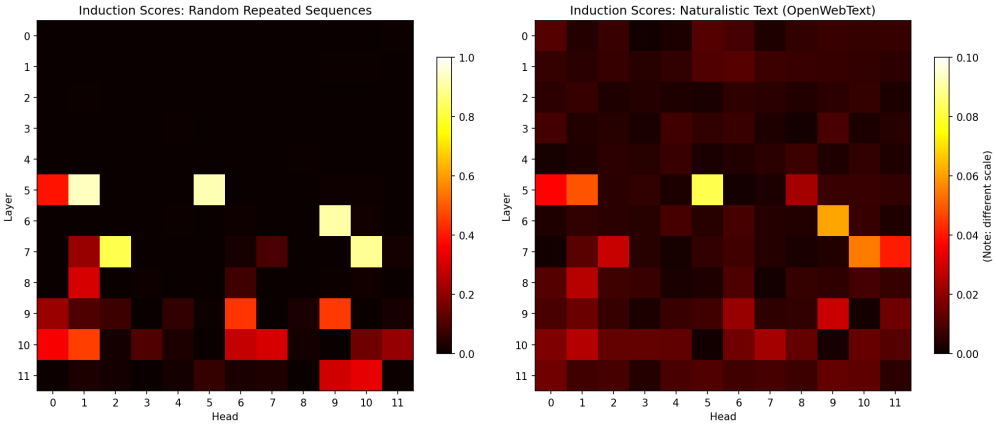

Figure 1: Induction score heatmaps across all 144 attention heads in GPT-2-SMALL. *(Left)*Random repeated sequences (scale 0–1). *(Right)*Natural text from OPENWEBTEXT (scale 0–0.1). The same heads score highly in both conditions, but natural-text scores are $\sim$10$\times$ lower. Note the different color scales.

A positive copying score indicates that the head's OV circuit tends to promote the token it attends to, consistent with a copying mechanism.

3.5 Ablation Study

To measure causal importance, we zero out attention patterns for specific head groups and measure the resulting increase in next-token prediction loss on natural text. We compare two groups: (1) all 21 heads with random induction score > 0.10 (the standard threshold), and (2) the top-5 heads ranked by naturalistic induction score.

3.6 Implementation Details

All experiments use PyTorch 2.10 on a single NVIDIA RTX A6000 GPU (49 GB VRAM). We use TRANSFORMERLENS for model access, hook-based activation caching, and attention pattern extraction. For the induction score threshold, we follow the standard cutoff of 0.10 used in prior work Olsson et al. [2022]. All random sequences use seed 42 for reproducibility.

4 Results

4.1 Induction Scores Are Highly Correlated but Dramatically Different in Magnitude

Figure 1 shows induction score heatmaps across all 144 heads (12 layers $\times$ 12 heads) on random and natural data. The spatial pattern of which heads score highly is similar across conditions, but the scales differ by an order of magnitude: the maximum random score is 0.937 (L5.H1) while the maximum natural score is 0.082 (L6.H9).

Figure 2 quantifies this relationship. The Spearman rank correlation between random and natural induction scores is $\rho = 0.839$ ($p < 10^{-30}$), and the Pearson correlation is $r = 0.826$ ($p < 10^{-30}$). All points fall well below the $y = x$ line, confirming that every head shows weaker induction behavior on natural text.

Table 1 lists the top induction heads and their scores in both conditions. The ratio between random and natural scores ranges from 11$\times$ to 40$\times$, with a median of 13$\times$ across the top five heads.

Top-K overlap. We also measure the overlap between the K highest-scoring heads in each condition (table 2). At $K = 5$, four of the top five heads are shared (80% overlap), and at $K = 20$, fourteen are shared (70%). Critically, no head crosses the standard 0.10 induction threshold on natural text—the maximum natural score is 0.082—yielding zero "universal" induction heads and zero "naturalistic-only" induction heads under the standard definition.

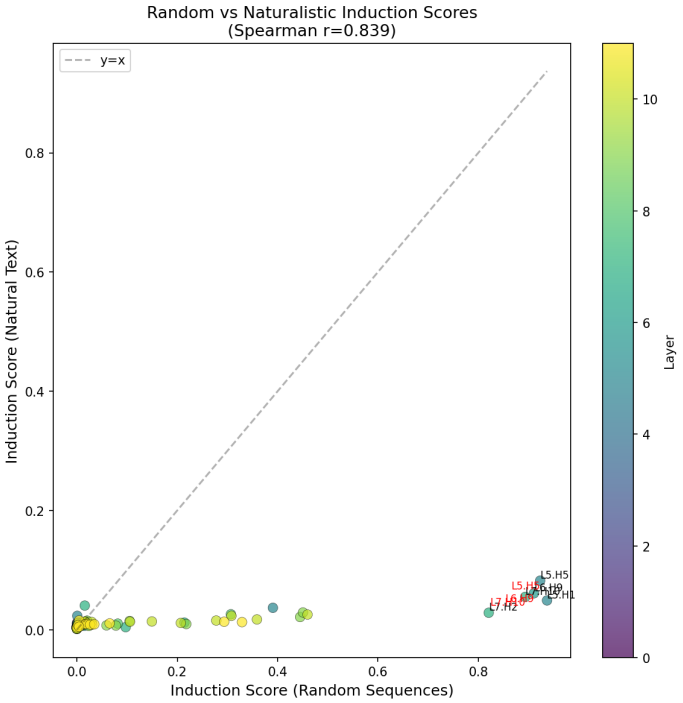

Figure 2: Scatter plot of induction scores on random (x-axis) vs. natural (y-axis) data for each of 144 heads. The strong correlation (Spearman $\rho = 0.84$) confirms that the same heads do induction in both conditions, while the distance from the $y = x$ line shows scores are 10–40$\times$ lower on natural text.

Table 1: Top induction heads ranked by random-sequence score. Natural-text scores are 10–40$\times$ lower despite similar token repeat rates (50% vs. 41%). Best natural score in **bold**.

Head	Random IS	Natural IS	Ratio
L5.H1	0.937	0.076	12.3$\times$
L6.H9	0.911	**0.082**	11.1$\times$
L7.H10	0.897	0.048	18.7$\times$
L7.H2	0.830	0.021	39.5$\times$
L5.H0	0.791	0.061	13.0$\times$

4.2 Fuzzy Matching Does Not Reveal Hidden Induction

One natural explanation for low naturalistic induction scores is that natural text requires *fuzzy* matching—attending to semantically similar rather than identical prior tokens. To test this, we computed fuzzy induction scores using cosine similarity > 0.8 between token embeddings.

The fuzzy scores are nearly identical to exact-match scores: the Spearman correlation between exact and fuzzy scores on natural text is $r = 0.990$, and the maximum fuzzy score (0.065) is actually *lower* than the maximum exact score (0.082). Relaxing the matching criterion does not reveal hidden induction behavior. The bottleneck is not matching precision but rather that natural text provides many useful attention targets beyond the induction-pattern position.

4.3 Late-Layer Heads Copy More on Natural Text

While the induction score measures where heads attend (the QK circuit), the OV copying score measures what they do with that information. Figure 3 reveals a striking pattern: late-layer heads (layers 9–11) show dramatically higher copying scores on natural text than on random data, even as their induction attention patterns are weaker.

Table 2: Overlap between top-K induction heads on random vs. natural data. High overlap confirms the same heads are identified in both conditions.

K	Shared Heads	Overlap (%)
5	4	**80**
10	7	70
20	14	70

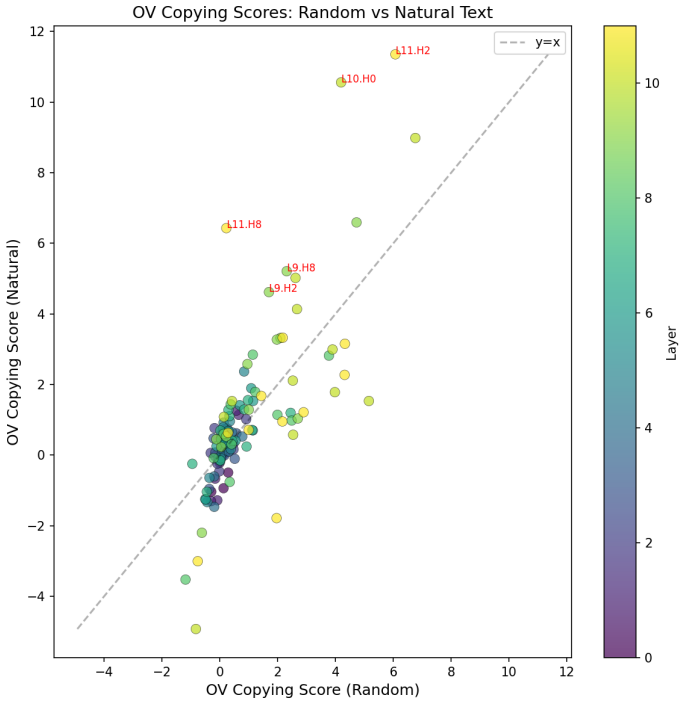

Figure 3: OV copying scores on random (x-axis) vs. natural (y-axis) data. Points above the diagonal indicate stronger copying on natural text. Late-layer heads (layers 9–11, warm colors) cluster above the diagonal, with L11.H8 showing a 29$\times$ increase on natural text.

Table 3 lists the heads with the largest naturalistic copying enrichment. Head L11.H8 increases its copying score from 0.22 on random data to 6.43 on natural text—a 29$\times$ jump. Head L10.H0 shows the largest absolute difference ($+6.37$). These heads' OV circuits are tuned for natural language patterns in a way that random token distributions fail to activate.

Figure 4 provides a layer-wise summary. Induction scores diverge sharply between conditions in layers 5–11 (random $\gg$ natural), while copying scores converge and even reverse in late layers (natural $>$ random). Attention entropy is similar across conditions, suggesting that the difference is not driven by more diffuse attention on natural text overall, but rather by a reallocation of attention away from the specific induction-pattern position to other contextually relevant positions.

4.4 Ablation Confirms Causal Importance

To verify that random-detected induction heads genuinely contribute to natural language processing, we zero out their attention patterns and measure the increase in next-token prediction loss on OpenWebText passages (figure 5).

Table 4 summarizes the results. Ablating all 21 random-detected heads increases loss from 3.047 to 5.380 ($+76.6\%$). Even ablating only the 5 heads with the highest naturalistic induction scores increases loss by 46.4%, confirming that these heads are not random-sequence specialists—they play a genuine role in natural language modeling, despite their low induction scores on natural text.

Table 3: Heads with the strongest naturalistic copying enrichment. Natural copying scores exceed random scores by 2–29× for late-layer heads, indicating distribution-sensitive OV circuits. Largest difference in **bold**.

Head	Natural CS	Random CS	Difference
L10.H0	10.56	4.19	**+6.37**
L11.H8	6.43	0.22	+6.20
L11.H2	11.35	6.07	+5.28
L9.H2	4.62	1.70	+2.92
L9.H8	5.21	2.31	+2.90

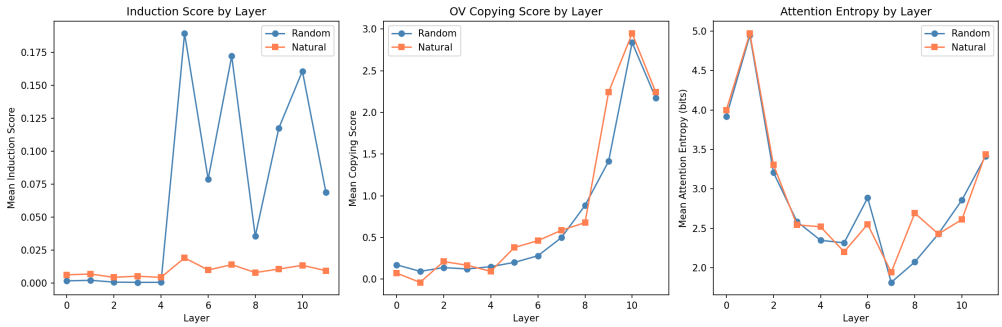

Figure 4: Layer-wise analysis of induction scores, copying scores, and attention entropy. *(Left)*Induction scores diverge sharply (random $\gg$ natural) in layers 5–11. *(Center)*Copying scores converge, with natural exceeding random in late layers. *(Right)*Attention entropy is similar across conditions, indicating the score difference is not due to globally more diffuse attention.

5 Discussion

5.1 The QK–OV Dissociation

Our most striking finding is the dissociation between attention patterns (QK circuit) and output behavior (OV circuit) across data conditions. On random data, the QK circuit produces strong induction attention patterns, and the OV circuit copies the attended token. On natural text, the QK circuit distributes attention across many contextually relevant positions—weakening the induction-specific signal—but the OV circuit actually copies *more effectively* when the attended content is natural language.

This dissociation suggests that the QK and OV circuits serve partially different functions depending on data distribution. The QK circuit in induction heads may function as a general "find relevant prior context" mechanism, of which the $[A][B] \ldots [A] \rightarrow B$ pattern is a special case that dominates on random data but is one of many useful patterns on natural text. Meanwhile, the OV circuit appears tuned to the statistics of natural language, producing stronger copying signals when the input matches the training distribution.

5.2 Why Are Induction Scores So Much Lower on Natural Text?

Three factors likely contribute to the 10–40× reduction in induction scores on natural text:

Attention competition. On random repeated sequences, the induction-pattern position is the single most informative attention target. On natural text, many positions carry useful information—syntactic heads, semantic associates, positional neighbors—creating competition that dilutes the induction-specific signal.

Multiple prediction strategies. On random data, copying is the only viable prediction strategy. On natural text, the model can leverage syntactic patterns, semantic associations, n-gram statistics, and many other cues, reducing reliance on any single mechanism.

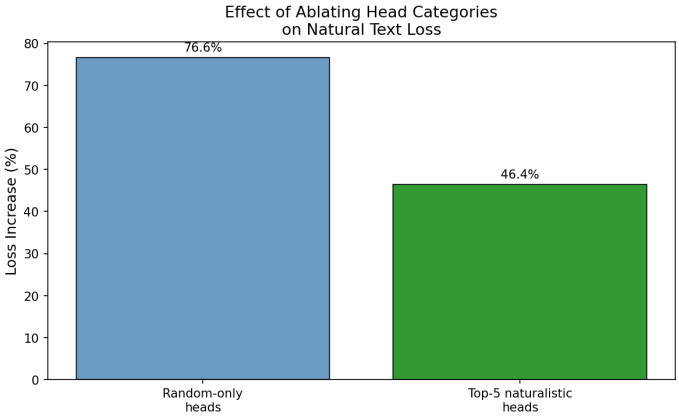

Figure 5: Ablation results on natural text. Removing the 21 random-detected induction heads (score > 0.10) increases loss by 76.6%, while removing only the top-5 naturalistic-scoring heads increases loss by 46.4%. These heads are not random-sequence specialists—they are causally important for natural language processing.

Table 4: Ablation results on OPENWEBTEXT passages. Both head groups substantially increase loss when removed, confirming their causal importance for natural language processing.

Ablated Group	N Heads	Ablated Loss	Loss Increase
None (baseline)	0	3.047	—
Random-detected (IS > 0.10)	21	5.380	**+76.6%**
Top-5 by natural IS	5	4.461	+46.4%

Distributional mismatch. The random-sequence setup, by forcing exact token repetition in a fixed pattern, creates a maximally favorable environment for the induction mechanism. Natural text, with its Zipfian token frequencies and rich contextual structure, is a far more complex prediction problem.

Notably, we ruled out one intuitive explanation: the difference is *not* primarily driven by fewer repeated tokens in natural text. Natural text has 41% repeated tokens versus 50% by construction in random sequences—a modest gap that cannot explain the 10–40$\times$ score difference. We also ruled out matching precision: fuzzy (embedding-similarity) matching does not recover higher scores, confirming that the bottleneck is attention competition, not token-identity constraints.

5.3 Implications for Mechanistic Interpretability

Our results have practical implications for how the field studies induction heads and, more broadly, how we evaluate circuit-level mechanisms:

Detection is reliable; quantification is not. Random-sequence detection correctly identifies *which* heads perform induction (high rank correlation, high top-K overlap). However, the *magnitude* of the induction score on random data dramatically overstates the strength of the induction attention pattern on natural text. Researchers should complement random-sequence detection with naturalistic evaluation when making claims about the importance of specific heads.

QK and OV circuits should be evaluated separately. The standard induction score conflates attention patterns and output behavior. Our results show these can diverge: a head may show weak induction attention on natural text while having a strongly distribution-sensitive OV copying circuit. Evaluating QK and OV contributions independently, as advocated by Elhage et al. [2021], provides a more complete picture.

Distribution matters for circuit evaluation. More broadly, our findings caution against evaluating circuits exclusively on synthetic data. The behavior of a circuit on artificial inputs may differ qualitatively from its behavior on the data distribution the model was trained on.

5.4 Limitations

Single model. We study only GPT-2-SMALL (124M parameters). Larger models with more layers and heads may develop more specialized induction mechanisms, including genuinely separate naturalistic-only heads. Scaling this analysis to GPT-2-medium, GPT-2-large, and the Pythia model suite is an important next step.

Single detection metric. The induction score measures one specific attention pattern. Other induction-like behaviors—attending to semantically similar contexts Ren et al. [2024], computing conditional probabilities Edelman et al. [2024], or performing abstract pattern completion—may require different detection methods that we do not explore here.

Short sequences. Our 257-token sequences are relatively short. Longer contexts may activate different attention patterns, particularly for late-layer heads that might benefit from richer context.

Static analysis. We analyze a fully trained model at a single checkpoint. The relationship between random and naturalistic induction behavior may evolve during training, with early-training heads potentially showing different cross-condition patterns than late-training heads Olsson et al. [2022], Singh et al. [2024].

Threshold sensitivity. The 0.10 induction threshold is conventional but somewhat arbitrary. Using percentile-based thresholds (top 10% of heads) would yield slightly different categorizations, though our correlation-based analyses are threshold-independent.

6 Conclusion

We conducted a systematic comparison of induction head behavior on random versus naturalistic data in GPT-2-SMALL. Our results show that induction heads are *not* data-type-specific: the same heads identified by random-sequence detection also show the strongest induction signals on natural text (Spearman $\rho = 0.84$). However, the nature of their operation differs dramatically across conditions. Induction attention patterns are 10–40$\times$ weaker on natural text, while late-layer OV copying circuits are up to 29$\times$ *stronger*—revealing a distribution-sensitive form of pattern completion that the standard detection protocol underestimates.

The answer to our motivating question—"are there hidden induction heads that only activate on naturalistic data?"—is: not as separate heads, but as separate *behaviors* of the same heads. Standard random-sequence detection identifies the right circuits but provides an incomplete picture of how they function on the data these models were trained to process.

Future work should extend this analysis to larger models, examine whether naturalistic-only induction heads emerge with scale, track how the QK–OV dissociation evolves during training using Pythia checkpoints, and connect our OV copying findings to the semantic induction mechanisms described by Ren et al. [2024].

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